

Detailed Syllabus Outline of DoSS Methods of Computing Bootcamp

Benjamin Smith

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Prior to Bootcamp

Survey

A survey has been sent out to assess the computational skills of students prior to the bootcamp. The questions are:

1. Your program is:

- ☐ MSc
- ☐ PhD

2. Indicate your highest level of completed degree

- ☐ Bachelors
- ☐ Masters

3. What did you major in?/ What was your concentration?

- (Open ended question)

4. Do you have experience with Git/Github

- ☐ No Experience
- ☐ Some Experience
- ☐ Advanced Experience

5. Do you have coding experience with R?

- ☐ No Experience
- ☐ Some Experience
- ☐ Advanced Experience

6. Do you have experience with the **tidyverse** library and relevant packages (e.g. dplyr, tidyr, ggplot2)?

- ☐ No Experience
- ☐ Some Experience
- ☐ Advanced Experience

7. Do you have experience with RMarkdown/Quarto

- ☐ No Experience
- ☐ Some Experience
- ☐ Advanced Experience

Module 0 - R Basics

Prior to the Bootcamp Starting, Module 0 will be posted with slides relevant to R basics. In particular the topics covered here will be:

1. Instructions to Download R and the RStudio IDE
2. How to Navigate RStudio
3. Setting a Working Directory
4. Installing an R package
5. Vectors
6. Matrices (Add matrix operations as well)
7. Data Frames
8. Lists
9. Running Linear Regression (no detail, just model fitting with `lm` and outputting with `summary()`)
10. Plotting with Base R
11. Some important math functions such as `floor()`, `ceiling()`, random variables generation (i.e. `rnorm()`, `rexp()`, `rbinom()`).
12. Writing functions
13. If/else
14. `for` loops
15. `while` loops
16. `next`
17. `break`
18. `apply()` (include `sapply()` and `lapply()`)
19. `knitr`- RMarkdown and Quarto
20. LaTeX basics.
21. Reading Data in R.
22. `head()`
23. The pipe operator (`|>` and `%>%`)
24. Resources (books)

Bootcamp Core Modules

Module 1 - Tidyverse Basics

1. Long and Wide form Data Definitions
2. `pivot_longer()` and `pivot_wider()`
3. Transforming data:
 - `select()`
 - `filter()`
 - `arrange()`
 - `mutate()`
 - `summarise()`
 - `group_by()`
 - `case_when()`
 - `ungroup()`
4. Plotting with `ggplot2`
 - Mention `ggsci` and `ggthemes`

5. Example of Plotting with `ggplot2` (`palmerpenguins` example)

- Scatterplot
- Histogram
- Boxplots
- Barcharts

6. Using LLMs to create Data Cleaning Scripts.

- Overview of major LLMs
 - ChatGPT
 - Gemeni
 - Claude
 - Grok
- Do not give LLM access to sensitive data!
- While using AI is inevitable, human contributions such as problem framing, key design decisions and thoughtful choices that make a data analysis meaningful are irreplaceable.
- Formulating a prompt to construct a data wrangling script/make a visual look nicer.

7. Exercises:

- Only if there is time
- Quiet work for a few minutes followed by sharing solutions.
- If not - post solutions after class or the next day

Module 2 - Statistical Inference (I)

1. Probability basics

- Cumulative Distribution Function (CDF)
- Probability Mass Function (PMF), Probability Density Function (PDF)
- Expectation & Variance Definitions
- Discrete R.Vs
 - Bernoulli & Binomial R.Vs
 - How to generate RVs in R
 - * `dbinom()`
 - * `pbinom()`
 - * `qbinom()`
 - * `rbinom()`
 - Example of Binomial Distribution Computing
- Continuous RVs.
 - Normal Distribution Examples
 - How to Generate them in R:
 - * Will provide a briefer description
- General overview of all distributions

2. Empirical and Theoretical CDF

3. Relationship between Probabilistic inference vs Statistical Inference.

4. Point Estimator Definition

- I would change this to the “Definition of Statistics”

5. MSE
6. Bias/Variance Decomposition of MSE
7. Unbiasedness
8. Consistency
9. Asymptotic Unbiasedness
10. Maximum Likelihood estimation
11. Exercise to Calculate MLE
12. Asymptotics of MLE
13. Cramer-Rao Lower Bound.
14. Calculating the MLE via

- `optim()`
- Newton Raphson

15. Exercises:

- Only if there is time
- Quiet work for a few minutes followed by sharing solutions.
- If not - post solutions after class or the next day

Module 3 - Statistical Inference (II)

1. Hypothesis Testing
2. Rejection Region
3. Type 1 error and Type 2 errors
4. Power Function ($\beta(\theta)$) and Size of Test (α)

- Shorten these definitions down to bullet points

5. Exercise on Power Function and Size of Test
6. Asymptotic Normality
7. Central Limit Theorem
8. Example of CLT
9. Likelihood Ratio Test.
10. Exercises:

- Only if there is time
- Quiet work for a few minutes followed by sharing solutions.
- If not - post solutions after class or the next day

Module 4 - Linear Regression/GLMs

1. Linear Regression
2. Interpretation of β_j coefficients.
3. Least Squares Estimation
4. Visualization of Least Squares Estimation (Projection of Y onto X)

5. Assumptions around ordinary least squares.
6. MLE vs OLS. GLMs
7. Asymptotics of MLE.
8. Logistic Regression
9. Exponential Family

- Gaussian
- Poisson
- Binomial

10. Link Function

- Specification in R

11. Exercises:

- Only if there is time
- Quiet work for a few minutes followed by sharing solutions.
- If not - post solutions after class or the next day

Module 5 - Simulation Studies and Parallel Computing

NOTE: The content here is less essential for students so it may be omitted if material in prior modules has not been completed.

1. Simulation Studies

- What is it
- Monte Carlo Simulation Definition
- Why perform simulation

2. Simulations for properties of Estimators

- Relative Efficiency Definition

3. Simple Examples

- Time difference
- Assessing Bias
- Sample Variance
- Relative Efficiency

4. Using lapply and vectorizing

- tradeoff between using a for loop vs vectorizing:
 - CPU load vs RAM load
 - (vectorizing uses more RAM, for loop uses more CPU)

5. Parallel Computing

- `parallel` package
- `doParallel` package
- `mclapply()`
- `parLapply()`

6. Error Handling

- `tryCatch()`

Supplementary Materials

Supplement 1

- Cover Git and Github Basics.

Supplement 5.

- Discussion of how to use HPC Cluster
- Discussion on Bootstrap